Remind: For A some real, square matrix we say A is orthogonally diagonalizable if there are real matrices P and D of same size as A such that

$$A = PDP^{-1}$$

where P is an orthogonal matrix and D a diagonal matrix.

Same as regular diagonalization, just additional condition that P is an orthogonal matrix rather than being any invertible matrix (which is a very extreme condition as orthogonal matrix is quite special!).

Showed last time that any orthogonally diagonalizable matrix is symmetric.

Turns out reverse is true - any symmetric matrix is orthogonally diagonalizable - very important result!

This is somewhat logical as we showed already that a symmetric matrix has only real eigenvalues and that eigenvectors which correspond to distinct eigenvalues are automatically orthogonal.

These two things are almost enough to make any symmetric matrix orthogonally diagonalizable. The main potential complication is same thing we initially worried about in regular diagonalization, having repeated real eigenvalue(s) without the proper number of linearly independent eigenvectors (i.e. some eigenvalue has its algebraic multiplicity not matching its geometric multiplicity).

Also, even if there are enough eigenvectors, are they orthogonal to each other? We showed eigenvectors corresponding to distinct eigenvalues are orthogonal in symmetric matrices, but what if some eigenvalue had two or more linearly independent eigenvectors corresponding to it? Not so clear then.

Fortunately, unlike with arbitrary square matrices, the former issue, repeated real eigenvalues, do not potentially prevent a symmetric matrix from being diagonalizable. As book says, this is tough to prove - we won't go through that, just assume it.

So assuming that, is it enough to make any symmetric matrix orthogonally diagonalizable? After all, the above only guarantees diagonalizable, not necessarily orthogonally diagonalizable.

Let's clarify the second issue to help see why this works. Again two eigenvectors which correspond to different eigenvalues are automatically orthogonal. (remember that's only because matrix is symmetric).

But a repeated eigenvalue will need multiple linearly independent eigenvectors corresponding to it, and those might not be inherently orthogonal to each other.

But can we make them so? Well in each eigenspace, those linearly independent eigenvectors are a basis. We can make them an orthogonal basis via Gram-Schmidt.

Then in each eigenspace, the linearly independent eigenvectors are orthogonal while two eigenvectors from different eigenspaces are automatically orthogonal so the full set we use is pairwise orthogonal which essentially gives us the needed orthogonal diagonalization. So this completes what we need.

So how to actually create the orthogonal diagonalization of a symmetric matrix? Actually mostly same process as regular diagonalization, just a couple of extra small nuances:

(1) If there are any eigenspaces of dimension higher than 1, the basis elements we take for that space must be an orthogonal basis (can turn it into that via Gram-Schmidt if needed).

(2) The eigenvectors we take must be normalized to have length 1 (easy to do, just divide each by its length).

7.1.15 Want orthogonal diagonalization of:

$$A = \begin{bmatrix} 3 & 4 \\ 4 & 9 \end{bmatrix}$$

---> $$A - \lambda I = \begin{bmatrix} 3-\lambda & 4 \\ 4 & 9-\lambda \end{bmatrix}$$

---> $$\det(A - \lambda I) = (3-\lambda)(9-\lambda) - 16 \stackrel{?}{=} 0$$

---> $$\lambda^2 - 12\lambda + 11 = 0$$

---> $$(\lambda - 11)(\lambda - 1) = 0 \text{ ----> } \lambda = 1, 11$$

So those are the eigenvalues. Now as usual, find the eigenvectors:

$$A - I = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}$$

so an eigenvector for this is: $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

And other one is:

$$A - 11I = \begin{bmatrix} -8 & 4 \\ 4 & -2 \end{bmatrix} \sim \begin{bmatrix} -8 & 4 \\ 0 & 0 \end{bmatrix}$$

here an eigenvector is: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

So regular diagonalization would give:

$$P = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \quad , \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 11 \end{bmatrix}$$

Since eigenvalues distinct, the eigenvectors are automatically orthogonal (can check also) so for orthogonal diagonalization we just need to normalize the columns of P giving us:

$$P = \frac{1}{\sqrt{5}}\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \quad , \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 11 \end{bmatrix}$$

And then since P is orthogonal:

$$P^{-1} = P^{T} = \frac{1}{\sqrt{5}}\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

Interestingly P and P^{-1} are the same here. Is that always the case? No because we could alter any column of P by multiplying it by -1 as that doesn't change its length. But it often will change whether P is symmetric (which is what is needed here to have $P = P^{-1} = P^{T}$).

7.1.17

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 5 & 1 \\ 5 & 1 & 1 \end{bmatrix} , \qquad \lambda = -4, 4, 7$$

Given eigenvalues since don't want problem to be too long. So we proceed to find eigenvectors:

$$A + 4I = \begin{bmatrix} 5 & 1 & 5 \\ 1 & 9 & 1 \\ 5 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 5 & 1 & 5 \\ 1 & 9 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{rl} - & 5x_1 + x_2 + 5x_3 = 0 \\ (& x_1 + 9x_2 + x_3 = 0 \;) \quad 5 \end{array}$$

---> $-44x_2 = 0$ ----> $x_2 = 0$

Then fill in first equation with a reasonable nonzero value, say $x_3 = 1$ ----> $x_1 = -1$

So $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = -4$

After that:

$$A - 4I = \begin{bmatrix} -3 & 1 & 5 \\ 1 & 1 & 1 \\ 5 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} -3 & 1 & 5 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

via noting R3 = -R1 + 2R2. Then we have:

$$\begin{array}{rl} - & -3x_1 + x_2 + 5x_3 = 0 \\ & (\ x_1 + x_2 + x_3 = 0\ ) \end{array}$$

----> $-4x_1 + 4x_3 = 0$

Then fill in a reasonable nonzero value to get say

$x_1 = 1$ ----> $x_3 = 1$ ----> $x_2 = -2$

So $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 4$. Then finally

$$A - 7I = \begin{bmatrix} -6 & 1 & 5 \\ 1 & -2 & 1 \\ 5 & 1 & -6 \end{bmatrix} \sim \begin{bmatrix} -6 & 1 & 5 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

via noting R3 = -R1 - R2. Yields:

$$+ \quad (-6x_1 + x_2 + 5x_3 = 0) \quad 2$$
$$x_1 - 2x_2 + x_3 = 0$$

----> $$-11x_1 + 11x_3 = 0$$

Fill in again a reasonable nonzero value to get say

$x_1 = 1$ ----> $x_3 = 1$ ----> $x_2 = 1$

So $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 7$.

Finally since eigenvalues distinct just normalize the eigenvectors to get the orthogonal diagonalization:

$$P = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

And then

$$P^{-1} = P^{T} = \begin{bmatrix} \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

Unlike regular diagonalization, in orthogonal diagonalization, you will usually be asked to find P^{-1}. Important to remember it's just equal to P^T, you don't have to do any additional calculation to find it!

Symmetric matrices have several important properties:

(1) Only real eigenvalues (proved in HW 5.5.27/28)
(2) Algebraic multiplicity of each eigenvalue is same as geometric multiplicity for each (hard to prove). This ensures diagonalizable over $\mathbb{R}$.
(3) Eigenspaces mutually orthogonal (proved in class)
(4) Based on prior three, every symmetric matrix is orthogonally diagonalizable.

Put together these are called the <u>Spectral Theorem</u>

Somewhat related, symmetric matrices break down in an interesting way. If A is a real symmetric $n \times n$ matrix and $\lambda_1, \lambda_2, \ldots, \lambda_n$ are its eigenvalues (all real but not necessarily distinct) with $u_1, u_2, \ldots, u_n$ the respective orthogonal eigenvectors, normalized (i.e. the columns of P) then:

$$A = \lambda_1 u_1 u_1^T + \lambda_1 u_2 u_2^T + \ldots + \lambda_1 u_n u_n^T$$

This is the <u>spectral decomposition</u> of the symmetric matrix.

Why is this true? Multiplying each side on right with u_1 this gives:

$$Au_1 = \lambda_1 u_1$$

as right side reduces to this since $u_1^T u_1 = [1]$ and u_1 is orthogonal to every other u_i.

And this statement is of course true since λ_1 is an eigenvalue of A with u_1 an associated eigenvector.

Doing this with each u_i will prove this. That is, A minus this proposed decomposition has each u_i in its null space. And since the u_i's span the associated $\mathbb{R}^n$ that matrix has nullity n and thus rank 0 so must be the 0 matrix, as desired.

Interesting concept which can be related to symmetric matrices that leads us to next section.

$$2x^2 + y^2 = 1 \quad : \quad \text{ellipse}$$
$$2x^2 - y^2 = 1 \quad : \quad \text{hyperbola}$$

Very similar looking equations, same exact style of terms, just one coefficient's sign changes. Yet the two curves are extremely different, why?

Also this one:

$$y = x^2 \qquad : \qquad \text{parabola}$$

Almost any curve in x, y which is of degree two (i.e. has at least one term of degree two and nothing of higher degree) is an ellipse, hyperbola, or parabola.

Those are called <u>conics</u>, something you probably studied in high school geometry.

What type of curve (conic) is:

$$x^2 + 3xy + 4y^2 + 5x - 2y = 100$$

Still only terms of degree two or less. But obviously a much more complicated looking curve.

You might have learned a test for this in high school geometry. But where did that test come from?

Turns out we can use ideas of symmetric matrices to help resolve this issue.

This is not an important topic in this class by any means it's just interesting to see how that topic, which started as a high school geometry problem, can be solved via via linear algebra means (i.e. symmetric matrices).

7.2 Quadratic Forms

For

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

consider

$$x^T x = x_1^2 + x_2^2 + \ldots + x_n^2 = \| x \|^2$$

Already seen this in norm/length/least squares.

We now generalize this: For A any fixed real $n \times n$ symmetric matrix define <u>quadratic form</u> $Q: \mathbb{R}^n \dashrightarrow \mathbb{R}$

which has

$$Q(x) = x^T A x$$

When $A = I$, that's the $x^T x$ mentioned earlier.

<u>Ex:</u> In $\mathbb{R}^2$ take

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

$$Q(x) = Q\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 3x_2 \end{bmatrix}$$

$$= x_1(x_1 + 2x_2) + x_2(2x_1 + 3x_2)$$

$$= x_1^2 + 4x_1 x_2 + 3x_2^2$$

This is a cross product (i.e. mixed) term.

This makes this very different than when $A = I$ as that didn't have that style of term. Sure the coefficients of the squared terms also have been altered, but generally that's much less of an issue.

For one, is this expression always positive, is it always negative, or can it take both positive and negative values (for nonzero x)?

Easy to answer that for $2x_1^2 + x_2^2$ or $2x_1^2 - x_2^2$

but the cross product term definitely complicates that issue for the example mentioned.

Ok, here not so hard since not many terms. But imagine there were four squared terms so that there might be six mixed terms (each pair), would be much harder to tell this then!

Why does A create mixed terms when identity did not?

Even though A is symmetric, it is not diagonal is key. Is there a way to get around that? That's next key thing, maybe diagonalization ideas can help? After all, symmetric matrices are always diagonalizable, in fact orthogonally diagonalizable, one reason why we might only do this for symmetric matrices.